



Original article

Real-time CFD in the simulation of aerial firefighting

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ABSTRACT

Aerial firefighting is crucial in the combat of wildfires, as the speed, range, and payload (water, fire-retardant, crew) advantages provided by aircraft enable much easier access to remote areas and faster response to active fires than ground vehicles. However, the challenging mission scopes and flight conditions, including low-altitude operations, smoke-induced low visibility, fire-induced turbulence, and complex terrain, encountered during these operations place the pilot and crew under high workload and at considerable risk of accidents. With the increased frequency and intensity of wildfires seen in recent years due to climate change and other factors, research into safer and more efficient aerial firefighting techniques, strategies, and training is essential. This paper presents a high-fidelity simulation environment, leveraging real-time computational fluid dynamics (CFD) in the loop and capable of evaluating different aerial platforms and firefighting techniques.

1. Introduction

Wildfires have emerged as a growing global threat, fuelled by a combination of human activities in wildland areas, climate change, and other environmental factors [1]. Recent years have seen a notable increase in the frequency, intensity and scale of wildfires, leading to significant destruction, loss of life, and substantial economic losses, such as the recent devastating wildfire that swept through the Los Angeles, San Diego and Ventura, California counties of the United States [2]. According to the Copernicus European Forest Fire Information System (EFFIS) [3], several European countries, such as Bulgaria, Portugal, and Greece, have seen a notable increase in the number of identified wildfires, with France, Italy and Romania seeing a doubling of their annual averages in 2025 alone. In the United Kingdom, the number of fires and burned area identified in 2025 has surpassed the previous record numbers of 2019 [3].

Aerial firefighting is a key asset in the combat of wildfires, especially in remote areas where ground-based firefighting teams struggle to access [4]. Aircraft deployed in firefighting missions have offered speed, range and payload advantages, enabling rapid delivery of crew, equipment, water, and fire-retardant to active fires and fire-prone locations [5]. However, these operations involve flights in hazardous conditions that require high levels of precision and skill from pilots and crew [1]. To effectively suppress wildfires, aircraft must fly at low altitudes while navigating through intense, unsteady winds and complex terrains [6].

The winds include atmospheric and fire-induced components, the later are generated by the intense fire heat, and also affects the handling of the aircraft. Moreover, aerial firefighting crews must contend with smoke generated by the wildfires [7]. The smoke significantly degrades the visual environment, which increases the pilot workload and can hamper situational awareness. Low-altitude manoeuvres under such flight conditions leave pilots with very small margins for error during operations, significantly increasing the risk of accidents.

These factors underscore the importance of accurate simulation tools and training environments for aerial firefighting. A realistic, controlled, and readily accessible training system will enable the effective evaluation and improvement of various aerial firefighting platforms and strategies, as well as safe and routine pilot training for such risky operations.

In this respect, there have been several attempts to develop aerial firefighting environments over the past decades. Many previous studies focused on improving the simulation realism by incorporating different fire models into existing flight simulation systems. Cistriani et al. [7] presented the development of a coupled fire-atmosphere model to be used in a real-time simulation of the Canadair CL-415 aircraft for flight training. The research coupled the Rothermel [8] fire spread model with a cellular automata-based propagation algorithm and incorporated the Coupled Map Lattice (CML) method to simulate the fire-induced wind and buoyancy effects. The system was designed with a functional goal of enhancing the perceived realism of an already existing flight simulator, and as such, did not include components needed to analyse the

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Nomenclature

Symbols

ρ	fluid density, kg/m^3
Ω	rotor angular speed, rad/s
α	thermal diffusivity, m^2/s
τ	density distribution relaxation time, s
τ_g	temperature distribution relaxation time, s
Δt	time step, s
ψ	rotor azimuth, rad
ξ	propagating flux ratio
ρ_p	fuel particle density, kg/m^3
ρ_b	oven-dry bulk density, kg/m^3
ϕ_w	wind correction factor
ϕ_s	slope correction factor
ϵ	effective heating number
R	rotor radius, m
T	temperature, K
A	rotor disk area, m^2
c_s	lattice speed of sound, m/s
g	gravitational acceleration, m/s^2
$\mathbf{F}$	external volume force, N/m^3
$\mathbf{u}$	velocity vector, m/s
ROS	fire rate of spread, m/min
Q_{ig}	heat of preignition, kJ/kg
M_f	fuel moisture fraction
r	radial coordinate, m
w	lattice weight function
I_R	reaction heat intensity, $\text{kJ}/\text{min}/\text{m}^2$
u, v, w	velocity vector components, m/s
x, y, z	position vector components, m

Acronyms

LBM	lattice Boltzmann method
SPH	smoothed particle hydrodynamics
LES	large eddy simulation
FDM	flight dynamics model
BEM	blade element method
CML	coupled map lattice
DLL	dynamic link library
CUDA	compute unified device architecture
HLSL	high level shader language
ADM	actuator disk method
AGL	above ground level

aircraft performance or firefighting effectiveness under the computed fire-induced turbulence. Another example of such a system is presented by Vuković et al. [9]. The work presented the coupling of a simplified atmospheric model to analyse the effect of temperature variations, due to the fire, on the predicted rotor power of a blade element method (BEM) model of the Mi-171Sh helicopter. However, the assumptions made in an attempt to reduce the complexity of the atmospheric model limited the extent of the aerodynamic results obtained from the developed system.

More recently, Liu et al. [10] presented a new attempt at modelling an aerial firefighting simulation environment. At the core of their system was a physics-based fire spread model, which provides a relatively detailed analysis of the combustion processes and underlying heat transfer, coupled with a raster, cellular automata propagation system. The atmosphere was modelled with the incompressible Navier-Stokes equations via a semi-Lagrangian solver with a pressure-Poisson treatment based on the method proposed by Stam [11]. The work opted for the less numerically accurate [12] but faster vorticity confinement method [13], to compensate for the numerical dissipation and conservation errors resulting from the first-order interpolation used in the selected

solver. To conduct human-not-in-loop tests on the developed simulation and the generic BET-based fixed and rotary wing mathematical models, a proportional-integral-derivative (PID) based automatic flight control system capable of waypoint navigation was developed.

A common limitation of these previous works lies in the choice of atmospheric model. High-fidelity Navier-Stokes solvers are generally too computationally expensive for real-time applications, and, as such, the cited studies have opted for simplified formulations to meet the real-time simulation constraint. This work implements an atmospheric model based on the Lattice-Boltzmann method (LBM), which has been shown in the literature to provide sufficient numerical accuracy [14,15] while maintaining a relatively high computational efficiency advantage over direct Navier-Stokes solvers. To further improve the accuracy and realism of the simulation environment, this work expands on the idea of coupled fire-atmosphere models by incorporating results from the flight dynamics model within the LBM atmosphere, allowing for a full two-way coupling between the fire, aircraft and atmosphere.

Another major limitation in these previous simulation works is the lack of a dedicated water or fire-retardant component that accurately models the interaction between the suppression agent, the fire and the aircraft. This omission is significant because the fire suppression action directly alters the fire intensity and the generated updraft in the drop zone, which in turn affects the computed forces and moments on the aircraft. Furthermore, the presence of the water and the required storage component, such as bambi baskets and external tanks, not only affects the weight and centre of gravity of the aircraft but also significantly degrades its aerodynamic performance [1,4]. The present work aims to fill this gap by integrating a high-fidelity water simulation module based on smoothed particle hydrodynamics (SPH) to track water or retardant spread and model the fire suppression effectiveness. Additionally, the interaction between the fire suppression agent and the fire will be fed back to the atmospheric model, creating a fully coupled system where the aircraft influences the fire, and the fire in turn alters the atmospheric conditions around the aircraft. This level of detail is essential for the system to be usable in pilot training and optimising firefighting strategies in high-risk environments.

This paper is organised as follows: Section II details the system architecture, its components, and the implemented coupling methods. Section III presents the results from the conducted demonstration exercises. Section IV provides the conclusions and outlines directions for future development.

2. Methodologies and tools

Fig. 1 presents the overall structure of the flight simulation framework developed in this work. The system consists of four coupled modules: an atmospheric model based on the lattice Boltzmann method, an elliptical fire propagation model, an aircraft dynamics model and a smoothed particle hydrodynamics-based fire suppression model. The modules exchange data at every step, with two-way communication ensuring consistent physical interaction between the components. The atmospheric model computes the velocity field as a function of the background wind, turbulence, and aircraft wakes. It receives heat fluxes from the fire model to drive the buoyant plumes and momentum sources from the aircraft model to represent the effect of the generated aerodynamic forces. In return, the LBM model provides local velocity disturbances to both the fire and aircraft modules. The aircraft interacts with the fire model via the SPH-based water or fire-retardant model, with the heat distribution output also used to generate the data needed for smoke visualisation within the aircraft cockpit. The technical details of each model are presented in the following subsections.

2.1. Fire spread and propagation model with terrain integration

Fire spread models are mathematical descriptions of fire ignition and spread, based on terrain and environmental factors such as fuel

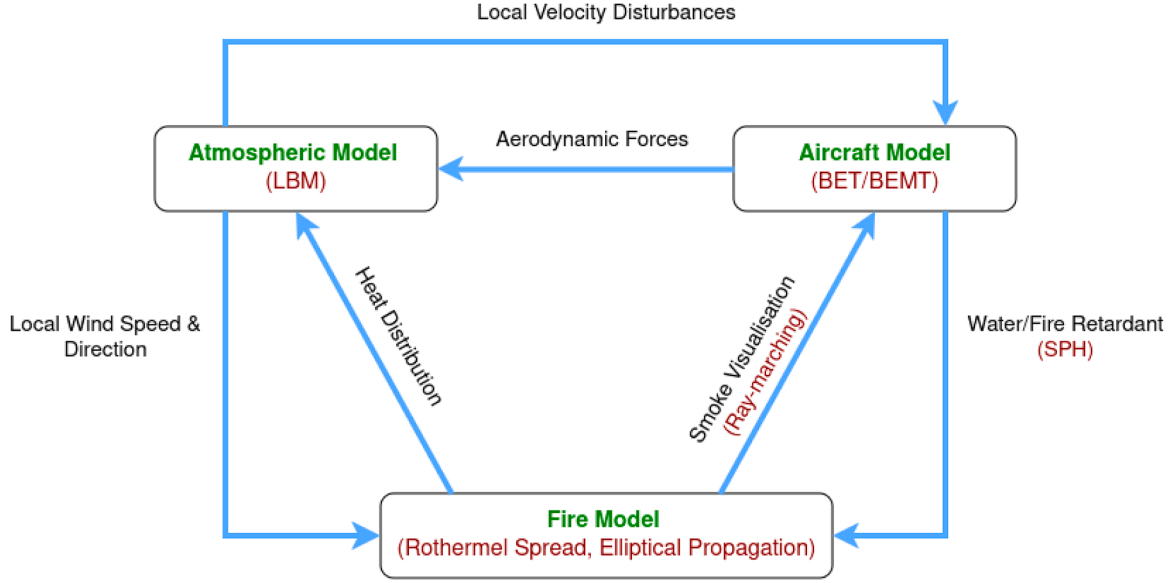


Fig. 1. Data flow between the components of the simulation showing the bi-directional interaction between components.

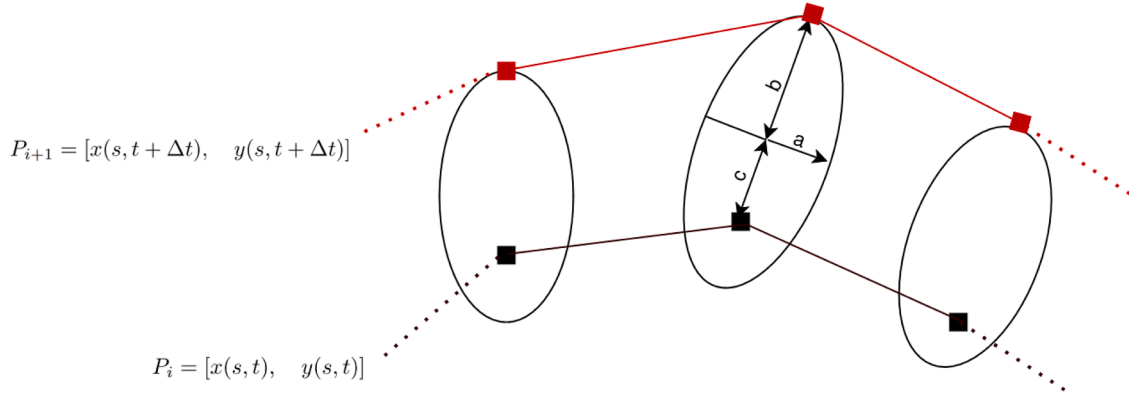


Fig. 2. Propagation of the fire perimeter, represented by the vertices (square points), between two time steps.

properties, ground moisture content, wind speed and direction, etc. From the literature, wildfire modelling methods used in research and industrial applications can be categorised into three types: statistical, physical, and empirical models. A comprehensive review of each model type is presented in references [16,17] and [18].

For this work, the semi-empirical Rothermel model [8] as defined in Andrews [19] was implemented. The model was chosen because it offers a good balance between prediction accuracy under varying environmental conditions, and computational efficiency that is crucial for real-time simulation. The rate of fire spread (ROS) output from the model is based on a given set of fuel and environmental variables and is given by:

$$ROS = \frac{I_R \xi (1 + \phi_w + \phi_s)}{\rho_b \epsilon Q_{ig}}, \quad (1)$$

where I_R is the reaction intensity, ϕ_w is the dimensionless wind factor, ϕ_s is the dimensionless slope factor, ρ_p is the fuel particle density, ϵ is the effective heating number, and Q_{ig} is the fuel heat of preignition.

To transform the computed one-dimensional spread rate into a two-dimensional form propagated across a terrain, an elliptical propagation method based on Huygens' expansion principle [20] was implemented. This involves representing the fire by a set of connected virtual ignition points, with an ellipse generated at each point based on the computed rate of spread. The perimeter of the fire at each timestep is then obtained by a curve that envelopes and connects the edges of the ellipses as shown in Fig. 2.

The shape of the fire ellipse at a given time t is determined by the head b , back c , and flank a , rates of spread, indicated in Fig. 2, and are calculated from the Rothermel model using the following expressions:

$$a(s, t) = ROS(s, t) \frac{1 + (LB - \sqrt{LB^2 - 1})^2}{2LB}, \quad (2)$$

$$b(s, t) = ROS(s, t) \frac{1 + (LB - \sqrt{LB^2 - 1})^2}{2}, \quad (3)$$

$$c(s, t) = b(s, t) - \frac{ROS(s, t)}{(LB + \sqrt{LB^2 - 1})^2}, \quad (4)$$

where the length-to-breadth ratio, LB , based on the prevailing mid-flame wind speed U_E , is given by 5 as defined in Anderson [21].

$$LB = 0.936^{0.2566U_E} + 0.461^{-0.1548U_E} - 0.397. \quad (5)$$

This study adopts the partial differential formulation for calculating the fire perimeter directly, provided by Richards [22], with the predictor-corrector integration method to obtain the position of the fire vertices at each timestep.

To integrate real-world landscapes into the simulation, a system that estimates the terrain slope, fuel properties, and height of each fire vertex from the input terrain data was implemented. The input data for

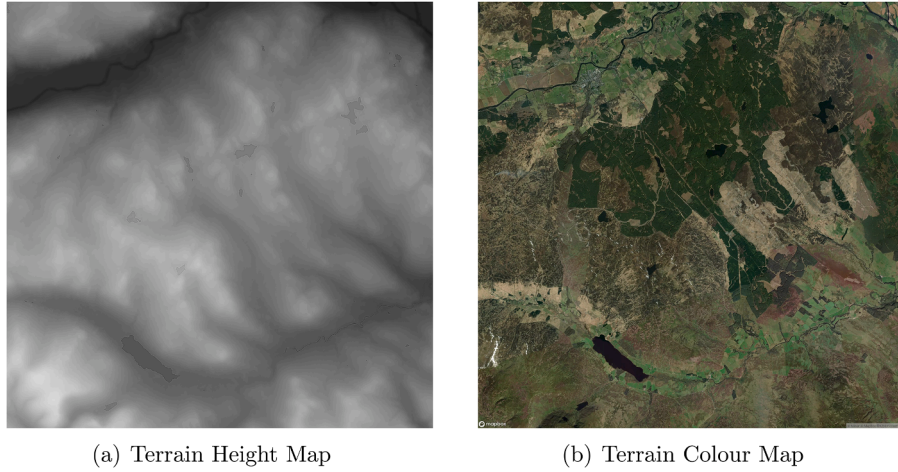


Fig. 3. Sample 18km by 18km height (a) and colour (b) maps from the Tay Forest Park (Latitude: 56.56939 deg, Longitude: -3.81882 deg) [23]. In the height map, the elevation is indicated by the greyscale brightness, with darker regions representing lower elevation areas.

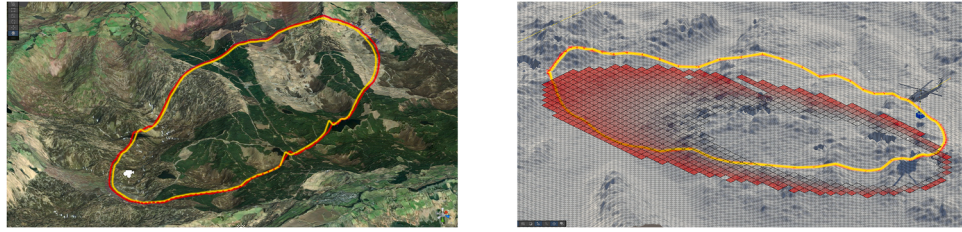


Fig. 4. Rendered view of the modelled fire spreading across the input terrain. The outer and inner fire perimeters are represented by the red and yellow lines, respectively. The right image shows the heat distribution of the terrain grid cells within the fire perimeter.

each terrain is a height texture map containing the normalised elevation, above sea level, for each grid point and a colour map which specifies the vegetation distribution across the landscape (Fig. 3).

By defining a cell size in meters, the maps can be discretised into an $N \times M$ grid with the fuel properties within each cell determined from a vegetation scale conversion, based on the evaluated terrain colour and height at each pixel. For example, a brown pixel at a relatively low region of the terrain is assumed to be made up of dry grass, while a similar colour at a high point in the terrain is assumed to be part of a rock structure.

For each vertex on the fire perimeter, the altitude, z , is determined by converting the vertex position from world coordinates into pixel coordinates and evaluating the normalised height at this position from the height map. The true height above the sea level for this point is then computed by rescaling the normalised height with the maximum height obtained within the real-world landscape. The terrain slope in the direction of movement for each fire vertex is obtained by the normalised velocity vector scaled by the specified grid size, which points towards where the vertex is moving. The height at this grid position, the height at the current grid position, and the distance between them are then used to compute the slope angle ϕ , whose tangent is fed into Eq. (6):

$$\Phi_s = 5.275\beta^{-0.3}(\tan\phi)^2, \quad (6)$$

to calculate the terrain slope needed in Eq. (1). Fig. 4 presents a view of the fire perimeter over the sample terrain and the representation of the temperature distribution within the area covered by the fire.

2.2. Water modelling and fire coupling

The suppression liquids (water and fire retardant) within the system are based on the smoothed particle hydrodynamics (SPH) method. This is a mesh-free, Lagrangian method in which the fluid is represented as

discrete particles, carrying properties such as mass, velocity, and density. These field quantities can be evaluated at any point within the simulation space by aggregating the properties of neighbouring particles, within a given space, with a radially symmetric smoothing kernel [24].

Lagrangian methods offer a number of advantages for fluid simulation. Firstly, the exact conservation of mass and momentum is ensured, given that the number of particles in the simulation and their masses are held constant. Additionally, and especially important for this work, the method avoids the need for computational grids, which simplifies the handling of complex boundaries and free surfaces, such as the terrain, making it ideal for real-time water modelling and simulation. An extensive overview of the method and its accompanying computation steps is given in Müller et al. [25] and Zhang et al. [26].

Within the algorithm, field quantities that are defined at discrete particle positions can be estimated by the weighted sum of each particle's contribution using:

$$A_S(\mathbf{x}_i) = \sum_j m_j \frac{A_j}{\rho_j} W(\mathbf{x}_{ij}, h), \quad (7)$$

where A_S is the scalar result at position $\mathbf{x}_i$ and j iterates over the neighbouring particles getting their mass m_j , density ρ_j and position $\mathbf{x}_{ij}$. The variable $W(\mathbf{x}, h)$ is a smoothing kernel of radius h that acts as an influence weighting factor, and determines how much the scalar value A_S is affected by the values of the particles within the neighbourhood.

The evolution of fluid field parameters such as density ρ , pressure p , and velocity $\mathbf{v}$ are defined by the Navier-Stokes conservation of mass and momentum equations for incompressible fluids as:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{\tilde{v}}) = 0, \quad \text{and} \quad (8)$$

$$\rho \left(\frac{\partial \mathbf{\tilde{v}}}{\partial t} + \mathbf{\tilde{v}} \cdot \nabla \mathbf{\tilde{v}} \right) = -\nabla p + \rho \mathbf{g} + \mu \nabla^2 \mathbf{\tilde{v}} + \mathbf{\tilde{f}}_{ext}, \quad (9)$$

where μ is the fluid dynamic viscosity, $\mathbf{g}$ is the gravitational acceleration and $\vec{f}_{ext}$ is the sum of the external forces applied to the fluid per unit volume. By applying particle approximations to these set of equations, the following expression:

$$\vec{a} = \frac{d\vec{v}}{dt} = - \sum_j m_j \frac{p_j + p_i}{2\rho_j} \nabla W(\mathbf{x}_{ij}, h) + \mathbf{g} + \mu \sum_j m_j \frac{\mathbf{v}_j}{\rho_j} \nabla^2 W(\mathbf{x}_{ij}, h) + \frac{\vec{f}_{ext}}{\rho}, \quad (10)$$

which describes the acceleration of the fluid particles as a function of the pressure, gravity, viscosity and external forces applied to the particles, is obtained. Where p_i is the pressure computed using the equation of state for particle i while $\nabla W(\mathbf{x}_{ij}, h)$ and $\nabla^2 W(\mathbf{x}_{ij}, h)$ are the derivative and Laplacian of the selected kernel functions respectively. For density calculations, the spiky kernel 11, given by Desbrun and Gascuel [27] was employed. The kernel has a very strong gradient especially within small particle separation distances which helps to avoid the particles clustering together. For the other compute functions within the algorithm, the sixth-order kernel function 12 was used for its computational efficiency especially during distance calculations.

$$W_{spiky}(\mathbf{x}_{ij}, h) = \frac{15}{\pi h^6} \begin{cases} (h-r)^3 & 0 \leq r \leq h \\ 0 & r > h \end{cases} \quad (11)$$

$$W_{poly6}(\mathbf{x}_{ij}, h) = \frac{315}{64\pi h^9} \begin{cases} (h^2 - r^2)^3 & 0 \leq r \leq h \\ 0 & r > h \end{cases} \quad (12)$$

Because each of the contained particles represents a given mass fraction of the fluid, the amount of fluid distributed within a given area of the simulation space can be quantified and evaluated by aggregating the masses of the particles identified within the region of interest. This is another important factor for this project, as it allows accurate tracking of the dispersion and coverage of the water or fire-retardant dropped during simulation.

In the SPH solver boundary resolution function, particles along the terrain edges are identified, and a callback detailing their position in world space, velocity and mass is issued. Based on the position at contact, the terrain grid cell within which the particle falls is identified, and the cell water mass, used in determining the moisture fraction, is increased using the mass of each particle within the cell. The moisture fraction is an input to the fire spread module, and a higher value results in a lower spread rate within this grid cell, essentially establishing an interaction loop with the fire whereby the fire spread is reduced or stopped due to the dropped water particles.

The sum of the particle masses within each grid cell is also used to estimate the temperature drop due to heat being transferred from the grid fuel to the dropped water during each update cycle. Given that the cell heat intensity outputs are the source terms within the LBM atmosphere heat distribution, this mass-based heat absorption provides a pathway for the interaction between the SPH water and the surrounding atmosphere.

2.3. Atmosphere model

2.3.1. Lattice Boltzmann method

To analyse the impact of the fire on the atmosphere around and above it, an atmosphere model based on the Lattice Boltzmann Method (LBM) was developed. The LBM is a mesoscopic computational approach that models fluid dynamics by simulating the statistical behaviour of fluid particle populations on a discrete lattice. Due to the parallel nature of its algorithm, the method has seen promising implementations on Graphics Processing Units (GPUs) [28–31] which have demonstrated how a real-time fluid simulation, with sufficient numerical accuracy, can be obtained by combining a GPU with an optimised Lattice Boltzmann solver.

The classic discretised lattice Boltzmann evolution equation with the selected collision operator and source term is given as:

$$f_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = f_i(\mathbf{x}, t) - \Omega(f_i) + S_{mi}(\mathbf{x}, t) \Delta t, \quad (13)$$

$$S_{mi}(\mathbf{x}, t) = \left(1 - \frac{\Delta t}{2\tau}\right) w_i \left[\frac{\mathbf{c}_i \cdot \vec{u}}{c_s^2} + \frac{(\mathbf{c}_i \cdot \vec{u})^2}{c_s^4} \right] \cdot \mathbf{F}(\mathbf{x}, t), \quad (14)$$

where f_i is the density distribution function, i is the current velocity direction, $\mathbf{c}_i$ is the lattice velocity vector along the discrete direction, w_i is the weighting factor, $c_s = 1/\sqrt{3}$ is the lattice speed of sound, S_{mi} is the momentum source term from the vector $\mathbf{F}(\mathbf{x}, t)$ applied to the lattice using the Guo et al. [32] forcing scheme. In 13, $\Omega(f_i)$ is the collision operator which relaxes the distribution function towards the local equilibrium and can be a selection of either the multiple-relaxation time (MRT) [33], the two-relaxation time (TRT) [34] or the most often used Bhatnagar, Gross and Krook (BGK) operator.

The macroscopic variables such as density ρ , and velocity $\vec{u}$, which is corrected for the applied force, are obtained from the moments of the distribution functions using Eqs. (15) and (16). Guo et al. [32] showed that this method can fully recover the exact Navier-Stokes equations as it captures the direct effect of the body force on the momentum and the effect of its temporal and spatial variations on the momentum and density.

$$\rho = \sum_{i=1}^Q f_i, \quad (15)$$

$$\vec{u} = \frac{1}{\rho} \sum_{i=1}^Q \mathbf{c}_i f_i + \frac{\Delta t}{2\rho} \mathbf{F}. \quad (16)$$

Within the simulation, the standard no-slip boundary condition is implemented for static walls, the zero-velocity gradient condition for the sides, and the generalised Zou and He condition, as described in Hecht and Harting [35], is applied to the inlet and outlet cells.

2.3.2. Thermal model and fire integration

The two-population LBM model, described in Guo et al. [36], was employed to incorporate the thermal effects of the fire within the simulation. Following the same principle as the density distribution, the temperature distribution function g_i using the BGK collision operator is given by the following expression;

$$g_i(\mathbf{x} + \mathbf{c}_i \Delta t, t + \Delta t) = g_i(\mathbf{x}, t) - \frac{1}{\tau_g} (g_i(\mathbf{x}, t) - g_i^{eq}(\mathbf{x}, t)) + S_{qi}(\mathbf{x}, t), \quad (17)$$

where τ_g is the relaxation time for the temperature distribution, which is related to the thermal diffusivity, α , by $\tau_g = 3\alpha + \frac{1}{2}$. The temperature equilibrium distribution function, g_i^{eq} is obtained in a similar fashion to the density distribution. The quadratic form of the equilibrium function is selected over the linear form for the temperature, to avoid a velocity-dependent diffusivity [37]. The resulting expression is given below:

$$g_i^{eq} = w_i T_i \left[1 + \frac{\mathbf{c}_i \cdot \vec{u}}{c_s^2} + \frac{1}{2} \frac{(\mathbf{c}_i \cdot \vec{u})^2}{c_s^4} - \frac{1}{2} \frac{\vec{u} \cdot \vec{u}}{c_s^2} \right]. \quad (18)$$

One of the outputs from the empirical fire model is the fireline heat intensity I_f , which defines the amount of heat released per unit time and length of the fire perimeter. A general expression for this term, according to Byram [38], is given as

$$I_f = H \cdot w \cdot ROS, \quad (19)$$

where H is the heat yield of the fuel, w is the weight of the fuel burnt, and ROS is the rate of fire spread. By defining an area covered by a section of the fire perimeter, the volumetric heat source, Q_i in W/m^3 can be evaluated from the fireline heat intensity and related to the heat source term in Eq. (17) by

$$S_{qi} = \left(1 - \frac{\Delta t}{2\tau_g}\right) w_i \frac{Q_i}{\rho c_p}, \quad (20)$$



(a) The helicopter equipped with an external belly tank. [39]



(b) The helicopter equipped with an accordion-style external tank. [40]

Fig. 5. The S-70 FireHawk helicopter dropping water from different tank configurations.

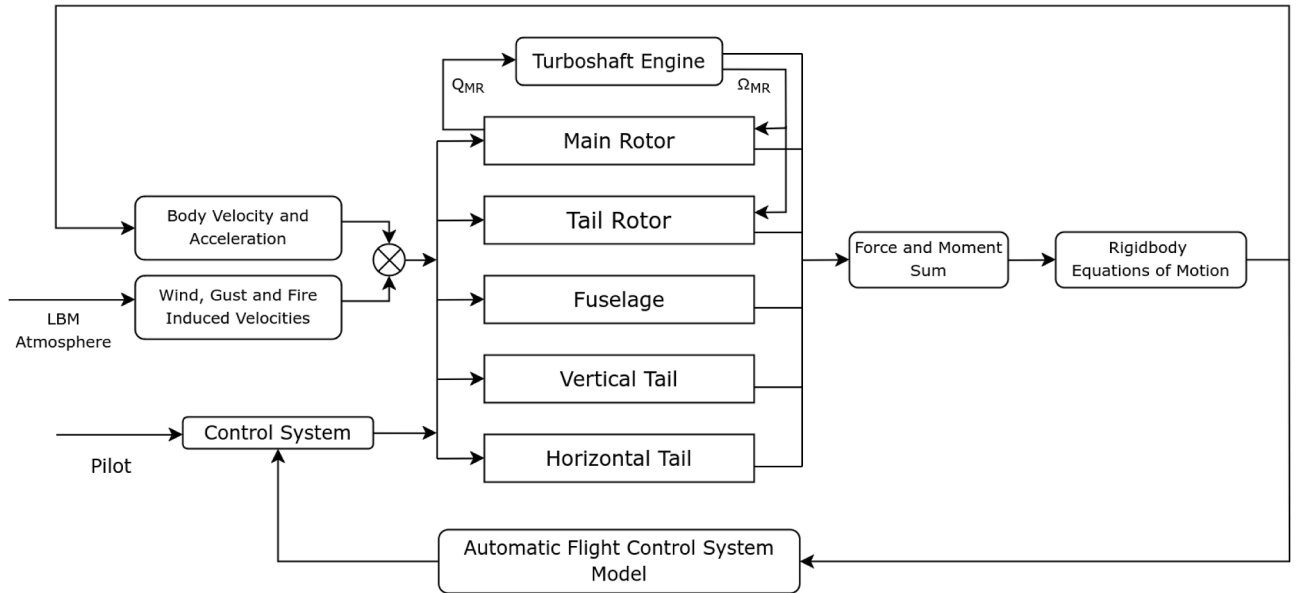


Fig. 6. GenHel flight dynamics simulation model structure.

where ρ is the air density and c_p is the specific heat capacity of air at constant pressure.

By taking the zeroth order moment, and including the effect of the spatial and temporal variation of the volumetric heat source, the temperature at a given lattice point can be calculated using Eq. (21).

$$T = \sum_{i=1}^Q g_i + \frac{\Delta r Q_i}{2\rho c_p} \quad (21)$$

The buoyancy force effect due to the fire-induced temperature changes is implemented based on the Boussinesq approximation, and the force applied to a given cell is proportional to the temperature difference between the current T_i and ambient state T_0 , as given by Eq. (22).

$$\vec{F}_b = -\alpha\rho_0(T_i - T_0) \cdot \vec{g} \quad (22)$$

2.4. Flight model

The Sikorsky S-70 FireHawk is a four-bladed, twin-engine, medium-sized transport and utility helicopter which can be reconfigured for search and rescue, medical transport and firefighting operations. The helicopter can be equipped with various water delivery methods such as the Bambi basket, the accordion style and conventional external belly tanks, as shown in Fig. 5(a) and (b) respectively [39,40]. This makes the helicopter a good candidate for this work, as a flight model developed

will not only serve as a means of evaluating the helicopter's performance itself, but also the effectiveness of different water delivery strategies.

The mathematical model implemented for demonstration purposes is a modification of the Sikorsky General Helicopter (GenHel) mathematical model, a modular simulation system designed to be a generalised mathematical model of a full helicopter system, described in references [41] and [42]. Fig. 6 shows an overview of the model, its components and data transfer between the components and from external sources. The main rotor aerodynamics is based on the Blade Element Momentum Theory (BEMT) whereby the blades are subdivided into sections with the angle of attack and aerodynamic forces and moments computed from the orthogonal velocity components and aerofoil tables. To compute the rotor's induced flow, the three-state Pitt-Peters dynamic inflow model [43] was incorporated into the developed system. The tail rotor model utilises the linearised closed-form Bailey rotor solution as defined in Bailey [44] and non-linear fuselage and empennage dynamics using look-up tables from wind tunnel tests formulated into equations which are functions of the local angle of attack, local sideslip, and dynamic pressure.

The engine power output is modelled using a full thermodynamic analysis of a two-spool turboshaft engine, following El-Sayed [45]. This represents the key components of the powerplant, including the intake, compressor, combustor, power turbine and the free turbine. This model will capture the effects of the changing atmospheric temperature and



(a) Motion Platform



(b) Simulator Cockpit

Fig. 7. Daedalus flight simulator showing the motion platform and the cockpit interior.

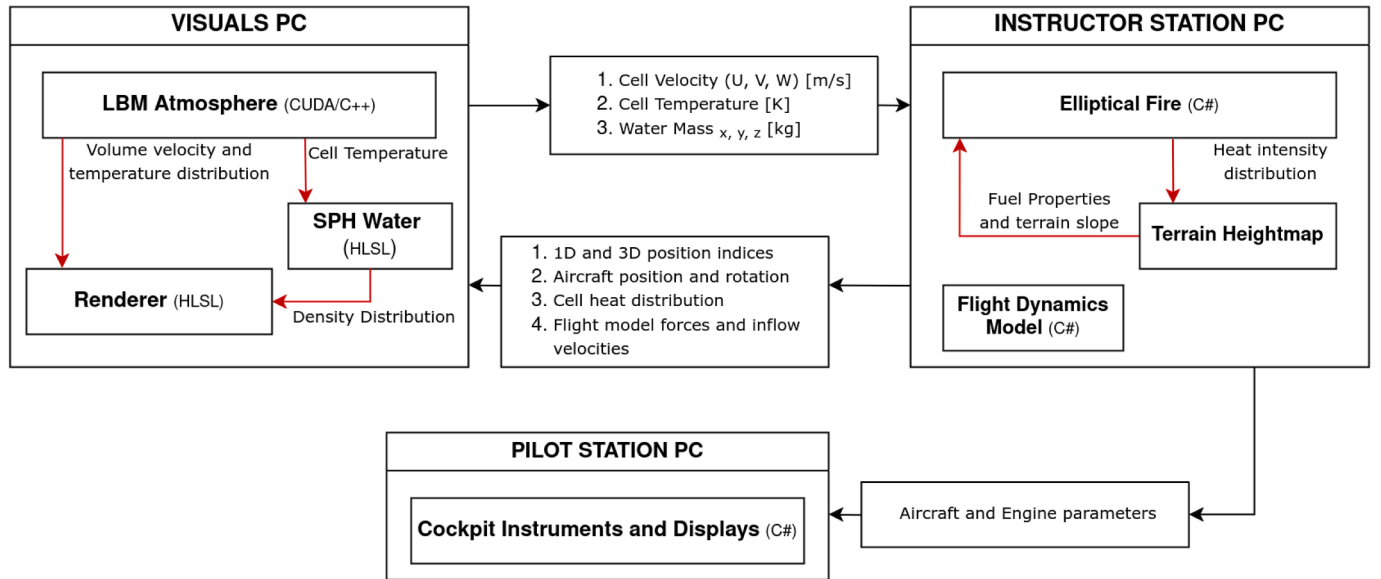


Fig. 8. Simulation system structure and component distribution across the underlying nodes.

density, as the aircraft operates in the vicinity of the fire, on the engine power and aircraft performance. To manage computational cost for a real-time application of the model, the rotational dynamics of the engine are approximated with a first-order lag model. This approximation provides an adequate representation of the time delay between a given throttle input and the engine response.

The system developed is a non-linear model with non-linearity in its components (e.g. the Navier Stokes equations) and the coupling between them, e.g. two-way exchange of information in the time-synchronised operational modes. Such a system needs careful examination and verification. Reference [46] put forward an interesting idea about symbolic computations and this is something to explore in the future. Furthermore, ideas for dealing with non-linearity are also put forward in Liu et al. [47] and again this is something to explore during the exploitation of the model in areas like fire-fighting training.

2.5. The daedalus I flight simulator framework and component coupling

The developed atmospheric, flight, and fire models were all integrated into the Daedalus I Flight Simulator framework of the University of Glasgow. The Daedalus I Flight Simulator, as shown in Fig. 7, is a high-fidelity flight simulation system built on a six-degree-of-freedom motion base, which allows it to realistically replicate the full range of aircraft movements, including pitch, roll, yaw, vertical motion, surge, and sway. The simulator utilises three high-power projectors that cre-

ate a panoramic, high-resolution visual environment within a cockpit approximating the Agusta-Westland AW-139 helicopter.

The cyclic, collective, and rudder pedals are fitted with BRUNNER [48] force feedback systems that reproduce the loads experienced on these controls during flight. This force feedback provides pilots with essential tactile feedback, enabling them to feel the control forces and, in combination with the motion platform, accurately simulate the physical demands of manoeuvring in challenging conditions. This flight simulator serves as the primary research platform for running the flight model and conducting all associated tests.

2.5.1. Simulation system design

The simulation environment employs a distributed computing and multi-language architecture to avoid pooling high computation workloads on a single device while also leveraging tools best suited for the computation and integration tasks. An overview of the system is shown in Fig. 8, highlighting the compute nodes and the included simulation components in each node. The distribution rule is based on the data transfer requirement between each component: components transferring a significant amount of data between each other must be on the same node, and any inter-node transfer is limited to 1024 bytes per compute step.

The explicit and parallel nature of the LBM algorithm means that it can take advantage of parallel computing architectures such as Graphical Processing Units (GPUs). The LBM solver within this project was

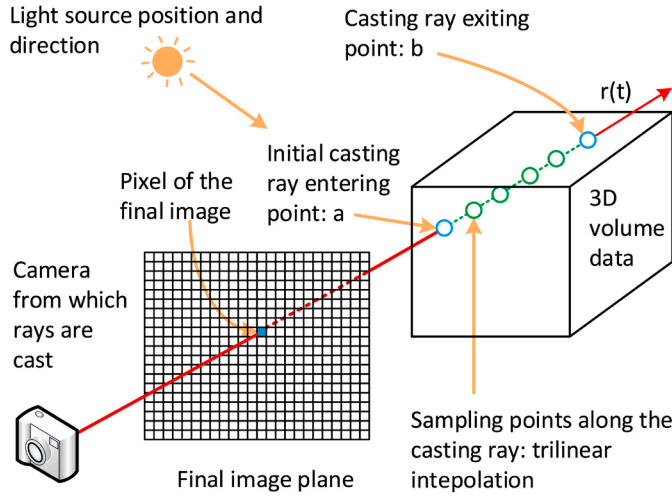


Fig. 9. Overview of the raymarching algorithm showing the ray start position, the entry point into the simulation volume, (a) and the exit point (b). The final colour and opacity of the pixel are specified as a function of properties sampled along the ray line [55].

implemented using the Compute Unified Device Architecture (CUDA) interface, a C++-like compute programming language for GPUs developed by Nvidia [49]. To facilitate a modular and easily extensible design, the CUDA-based LBM solver is implemented as a dynamic link library (DLL), which allows the solver to be initialised, updated and queried in real-time from the central simulation system irrespective of the language choice.

The atmospheric rendering and the SPH-based water simulation are implemented in the high-level shader language (HLSL), utilising GPU-accelerated compute and fragment shaders for fast parallel computation, while the other components, including the central system control, flight dynamics and fire propagation modules, are implemented in the C# language to efficiently integrate with the existing components within the Unity [50] development platform. To enable the large amount of data to be transferred between the simulation compute and visualisation components, a high-bandwidth, low-latency communication method is required. This project takes advantage of the interoperability functionality provided by the CUDA architecture with rendering interfaces such as DirectX9 [51], DirectX11 [52] and OpenGL ES [53]. This mechanism provides a way of sharing resources and data between processes running within these interfaces and a regular CUDA compute process, thereby avoiding the slow CPU-data transfer between device and host processes.

At the start of the simulation, the memory required to be shared between the components is allocated as compute buffers within the DirectX-based Unity environment. These buffers are first transformed into standard DirectX buffers before being converted into static array pointers for the LBM solver to interact with. With this connection, both the C#-based components and the LBM solver have access to the same memory location, and data written to this memory address is immediately available to the C# components without any transfer required.

2.5.2. Fire, smoke, and water visualisation

To visualise the fire, smoke and water within the simulator cockpit at runtime, the raymarching volumetric rendering algorithm was implemented. Within the scene, the positioning of the fluid is determined by the minimum and maximum position vectors of the simulation volume. The fluid is then rendered by marching through the volume with rays originating from the camera, and accumulating properties which determine the render colour and opacity of each pixel.

In the pixel fragment shader, a Ray-Plane intersection is performed for a ray originating from the pixel world position, in a direction pointing at the world camera position, using the slab method as described in

Majercik et al. [54]. This intersection logic checks if the ray intersects with the defined simulation volume, and if it does, returns the entry and exit positions as shown in Fig. 9 [55]. With these positions, the travel distance of the ray through the volume and the marching step size for each iteration step, based on the required number of steps (a value that controls the level of rendering detail), can be determined.

In the raymarching loop, a ray begins at the defined entry position and moves towards the exit point at the calculated step size. At each step position, a trilinear interpolation is performed to evaluate the fluid properties at that position and how much light has been absorbed by the fluid using Beer-Lambert's law [23].

$$I = I_0 e^{-p\mu_e d} \quad (23)$$

Here, I_0 is the initial light intensity, p is the fluid property, d is the ray-marching step size, and μ_e is the extinction coefficient which determines how much light is absorbed by the fluid as a function of the property being evaluated. The sampled density or temperature is then used to determine the colour of the pixel, while the computed light intensity is used to determine the magnitude of the alpha channel.

2.5.3. Helicopter dynamics and Lattice Boltzmann coupling

In the fully coupled simulation mode, the lattice Boltzmann solver analyses the flow field within the domain and includes the effect of the fire and the induced velocity from the helicopter. The coupling method is based on computing the helicopter rotor inflow using the velocity sampled from the lattice Boltzmann flow-field results instead of the empirical method used in the non-coupled mode. With this, the rotor inflow can be better predicted while also allowing the helicopter to respond to changes within the simulation domain due to the presence of the fire.

During each simulation step, the flight dynamics module sends the aerodynamic control points, in world space, and the generated physical forces at those points to a transport component, which then converts the positions to indices within the lattice Boltzmann domain and the forces to lattice source terms before sending them to the solver. With the cached indices, the transport component issues a request for the current fluid properties at a specified index and stores them against the world positions, which can then be sent back to the flight module.

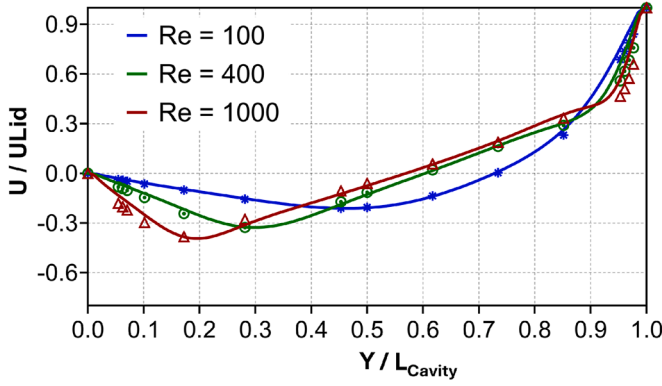
The current method for evaluating the rotor inflow is based on the work presented by Bludau et al. [56]. In a similar fashion to the three state Pitt-Peters inflow model, the velocity within the disk at a given radial position r and azimuth ψ can be evaluated from the mean $\mathbf{u}_o$, sine $\mathbf{u}_s$ and cosine $\mathbf{u}_c$ velocity components averaged from the identified cells within the disk using Eq. (24).

$$\mathbf{u}(r, \psi) = \mathbf{u}_o + \mathbf{u}_s \frac{r}{R} \sin(\psi) + \mathbf{u}_c \frac{r}{R} \cos(\psi). \quad (24)$$

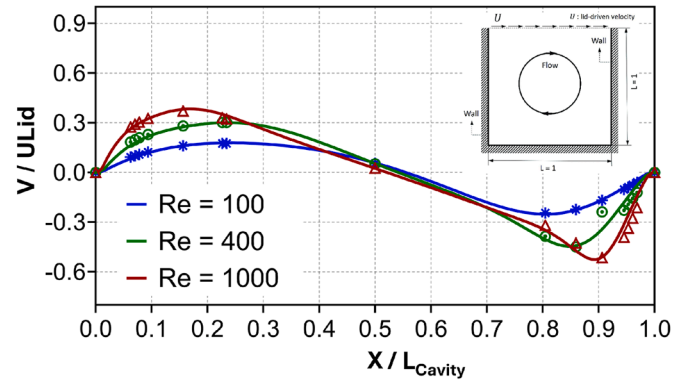
By first identifying the cells within the rotor disk, at its current location and orientation within the simulation domain, the velocity components can be requested from the transport component and then integrated to calculate the required inflow terms using the following expressions:

$$\begin{aligned} \mathbf{u}_o &= \frac{1}{\pi} \iint \mathbf{u} dA \approx \frac{1}{N_{disk}} \sum_{i=1}^{N_{disk}} \mathbf{u}_i, \\ \mathbf{u}_s &= \frac{4}{\pi} \iint \mathbf{u} \frac{r}{R} \sin(\psi) dA \approx \frac{4}{N_{disk}} \sum_{i=1}^{N_{disk}} \mathbf{u}_i \frac{r}{R} \sin(\psi), \\ \mathbf{u}_c &= \frac{4}{\pi} \iint \mathbf{u} \frac{r}{R} \cos(\psi) dA \approx \frac{4}{N_{disk}} \sum_{i=1}^{N_{disk}} \mathbf{u}_i \frac{r}{R} \cos(\psi), \end{aligned} \quad (25)$$

where r is the radial distance of the current cell from the rotor centre, ψ is the cell azimuth, $\mathbf{u}_i$ is the cell velocity, and N_{disk} is the total number of cells within the rotor disk bounds. Although computationally efficient, this method can be expanded upon to use non-averaged data and analyse the inflow through each blade to better capture finer velocity variations through the rotor disk.

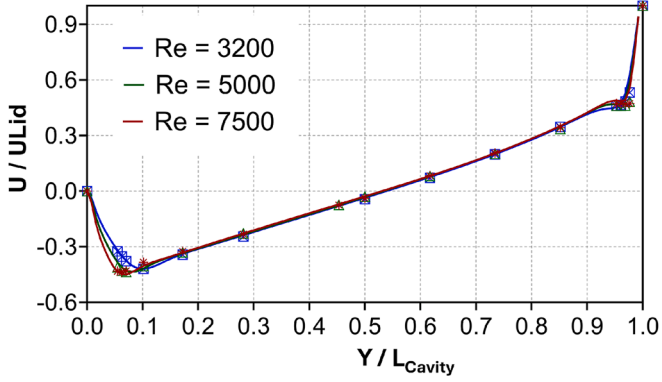


(a) Comparison of the x-velocity component along the cavity midpoint with experimental data.

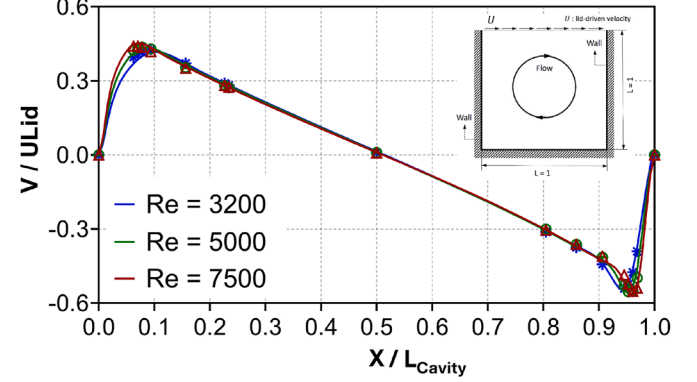


(b) Comparison of the y-velocity component along the cavity midpoint with experimental data.

Fig. 10. Comparison of the velocity profile of the LBM-BGK solver along the centre-line with results from literature.



(a) Comparison of the x-velocity component along the cavity midpoint with experimental data.



(b) Comparison of the y-velocity component along the cavity midpoint with experimental data.

Fig. 11. Comparison of the velocity profile of the LBM-MRT solver along the centre-line with results from literature.

To couple the forces generated by the helicopter main rotor, the actuator disk method (ADM) was employed within the LBM solver. The ADM applies a body force to the lattice cells within the rotor disk region to simulate the momentum exchange associated with thrust generation [57]. During each update step, the LBM receives the current rotor hub position and rotation, which then allows it to identify lattice cells inside the rotor disk that lie on the current rotor plane. The thrust generated by the rotor is then distributed over the disk, and the resulting volume force is applied alongside the buoyant force generated from the fire, and applied as a source term in the LBM forcing scheme using Eq. (14).

3. Results, demonstrations, and discussions

3.1. Validation and evaluation of the atmospheric model

To validate the accuracy and stability of the lattice Boltzmann implementation, two classic benchmark cases were simulated: the lid-driven flow and natural convection in square cavities. These cases provide individual tests of the solver's capability of simulating incompressible isothermal and thermal problems.

The lid-driven cavity test case consists of an incompressible flow within a square domain of length L , wherein the top wall moves at a

constant velocity U_{top} along the x-axis, while the remaining walls remain stationary with no-slip conditions. This setup induces a strong primary vortex and secondary corner vortices, and shows increasing unsteadiness up to the eventual transition to turbulence at higher Reynolds numbers. The results obtained from the current implementation were compared with established benchmark data from Ghia et al. [58], and the velocity profiles for $Re = 100, 400$ and 1000 using the standard single relaxation time BGK collision operator, Fig. 10, showed good agreement. For higher Reynolds numbers, the MRT collision operator was needed to ensure numerical stability within the simulation and the results obtained for Re values $3200, 5000$ and 7500 , Fig. 11, also agreed well with the results from literature.

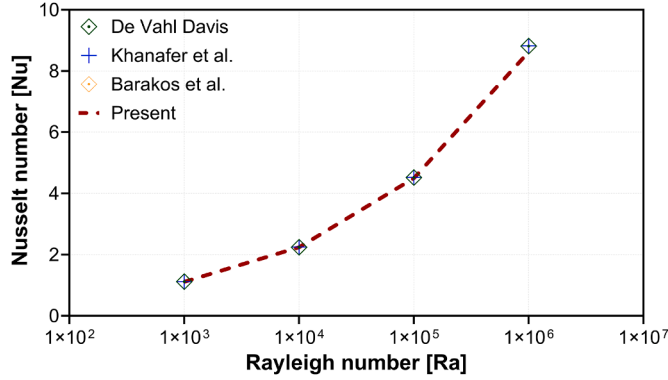
The natural convection case, on the other hand, involves a square cavity with differentially heated vertical walls with adiabatic boundary conditions along the other wall surfaces. The fluid motion in this case is driven by buoyancy, which is modelled using the Boussinesq approximation. This test is especially important as it allows the evaluation of the solver's ability to simulate thermal transport, which is essential for fire coupling.

To evaluate the accuracy of the thermal coupled LBM with the natural convection benchmark, the Nusselt Number, which gives the ratio between the total and conductive heat transfer at a given boundary,

Table 1

Brief overview of the various helicopter-LBM coupling modes and corresponding evaluation cases.

Case name	Rotor Inflow Source (Flow-to-aircraft coupling)	Rotor Force Feedback (Aircraft-to-flow coupling)
No-coupling	Uniform only	No
One-way A	Uniform Inflow Corrected with LBM	No
One-way B	Pitt-Peters Corrected with LBM	No
Full-coupling	LBM Inflow Only	Yes

**Fig. 12.** Comparison between the average Nusselt numbers at various Rayleigh numbers between the developed model and results from De Vahl Davis [59], Khanafer et al. [60] and Barakos et al. [61].

was calculated and compared with the corresponding results from the literature. The average Nusselt number was calculated on the left wall using Eq. (26):

$$Nu_{avg} = \frac{1}{T_{i-1} - T_{i+1}} \sum_{k=1}^{k=N} \frac{3T_{w,k} - 4T_{w+1,k} + T_{w+2,k}}{2\Delta x}, \quad (26)$$

where T_i is the temperature at a given index and Δx is the spacing within the lattice. Given the lattice size, H and the coefficient of thermal expansion, β , the Rayleigh number, Ra is defined as

$$Ra = \frac{g\beta H^3 (T_H - T_C)}{\alpha \nu}. \quad (27)$$

A comparison between the Nusselt number results obtained at various Rayleigh numbers from the model and similar results from the literature is shown in Table 1 and as seen in the table, there is a good agreement between the outputs (Fig. 12).

3.2. Validation and evaluation of the flight model

The helicopter was configured at a gross weight of 7257.5 kg, an altitude of 1600 m AGL and with the centre of gravity station-line, waterline positions at 351 and 231.5 in. respectively. A Newton-Raphson-based six-degree-of-freedom helicopter trimming algorithm was applied to the developed model to obtain its steady-state trim performance, and the results were compared against the flight test data [62]. The trim algorithm iteratively computes the state vectors and control inputs required to achieve the trimming target, i.e. level flight. The process begins by calculating the Jacobian matrix from the state F and control G matrices obtained with a central differencing scheme given as

$$F = \frac{\dot{x}_2 - \dot{x}_1}{2\Delta x}, \quad (28)$$

$$G = \frac{\dot{x}_2 - \dot{x}_1}{2\Delta u}.$$

The trim vector $\vec{A}$ combining the trim state and input components, at each timestep, is obtained by iteratively solving Eq. (29) until the computed deviation from the trim state derivative to the required state

derivative is within the specified tolerance value.

$$\vec{A}_i = \vec{A}_{i-1} - 0.5 * Jac^{-1} * [\dot{x}_i - \dot{x}_{target}] \quad (29)$$

where $Jac = [F, G]$ is the assembled Jacobian matrix, $\dot{x}_i$ is the state derivative at the current timestep and $\dot{x}_{target}$ is the required derivative vector.

Using this process, the aircraft was trimmed at various speeds ranging from 0 to 160 knots, and the resulting level flight trim control inputs, aircraft attitude, and main rotor power results were compared with the flight test data, and the good agreement between the two can be seen in Fig. 13.

3.3. Evaluations of system coupling

A couple of numerical experiments were performed to demonstrate the integration of the components within the simulation environment. The Lattice Boltzmann method-based atmospheric model used the D3Q19 discretisation on an open computational domain measuring 196 m along the X, Y and Z axes, representing 24 times the rotor radius along each direction. A diagram of the domain indicating the terrain, fire and helicopter positions is shown in Fig. 14. The LBM lattice size was set to 128 nodes along each axis, representing a spatial resolution of approximately 1.53 m per cell. This was a compromise between the required cell resolution along the rotor disk and the computation load resulting from the fact that the solver compute step is required to run about 15 to 20 times for each simulation step to keep the physical and lattice times matched.

The domain inlet and outlets used the Zou/He velocity and pressure boundary conditions, respectively, with the zero velocity and temperature gradient boundary enforced along the other axes except the ground ($y = 0$), where the no-slip condition was enforced. The inflow into the domain was designated with an atmospheric boundary layer velocity profile moving along the positive x-axis with a given reference velocity, U_{ref} , and reference height h_0 . Immersed boundaries were used to represent the helicopter fuselage, whereby the momentum within the marked cells was corrected using the current helicopter body velocity.

For all the simulation cases, the S-70 helicopter model with a gross mass of 7417 kg and a main rotor radius of 8.177 m was considered. The helicopter centre of gravity was set $x = -0.387m$, $y = 0$, $z = 0.209m$ in the helicopter local space, while the inertia tensor components were specified as $I_{xx} = 6316.76 \text{ kg/m}^2$, $I_{yy} = 6316.76 \text{ kg/m}^2$, $I_{zz} = 49888.7 \text{ kg/m}^2$ and $I_{xz} = 2551.65 \text{ kg/m}^2$ as given in Ballin [62]. The first mission involved the helicopter performing a level forward flight through the simulation domain containing a buoyant plume generated by the fire model. The objective was to evaluate the effect of the fire and the resulting atmospheric disturbances on the helicopter's performance during the flight. The helicopter, equipped with a rudimentary autopilot, maintained a constant altitude of 60 metres AGL and a speed of 20 knots, allowing for a controlled comparison between multiple coupling strategies.

Four distinct coupling modes were compared: The *No-coupling* case, where the uniform inflow model was used to compute the rotor forces and moments with no interaction with the CFD field, the *One-way coupling A* wherein the rotor inflow, evaluated from the LBM velocity field, was superimposed on a uniform inflow model to calculate the total inflow through the rotor, the *One-way coupling B* where the Pitt-Peters dynamic inflow was combined with the sampled CFD inflow, and the

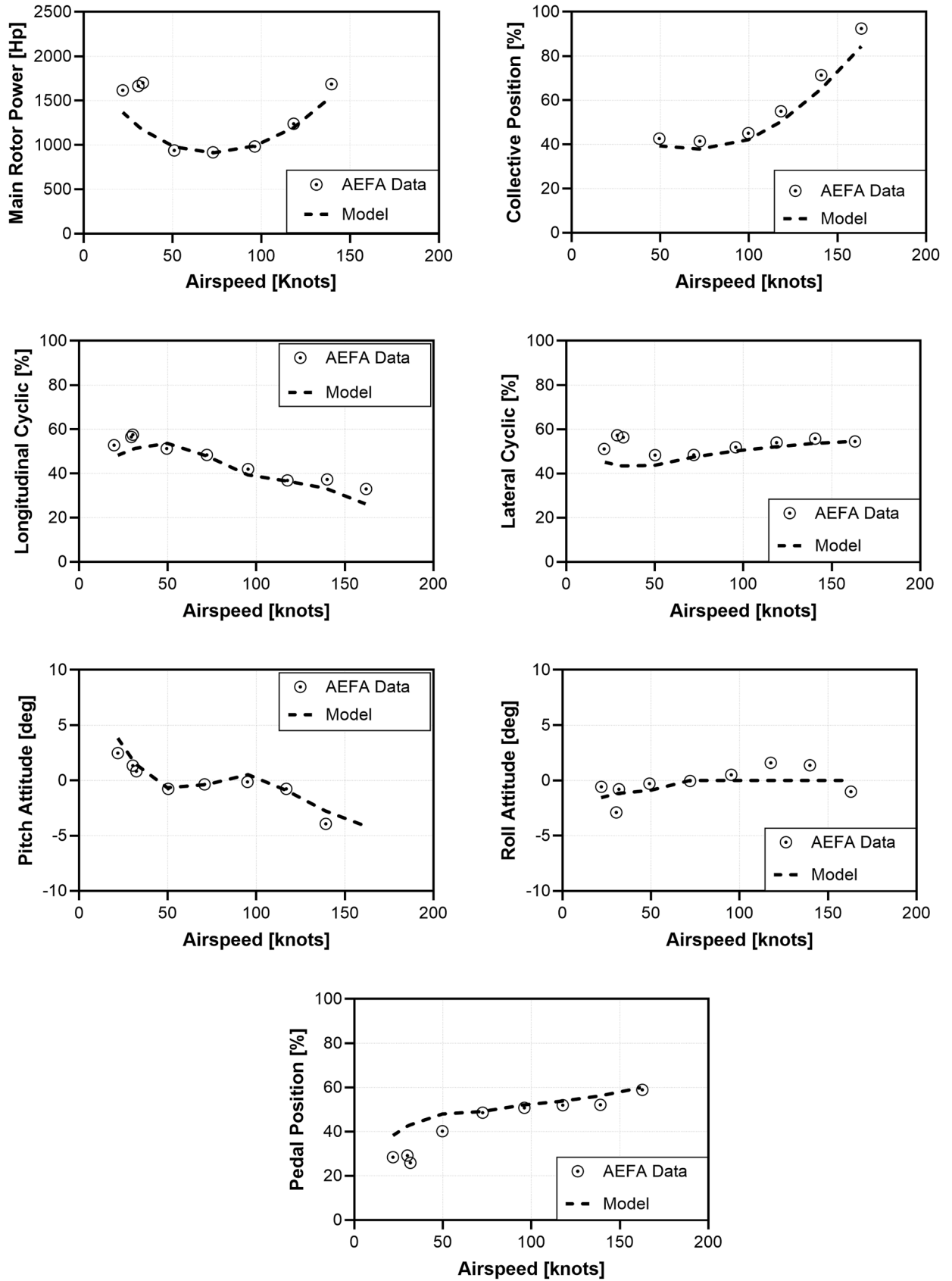


Fig. 13. Level-flight static trim results from the developed model compared with USAAEFA [62] flight test data.

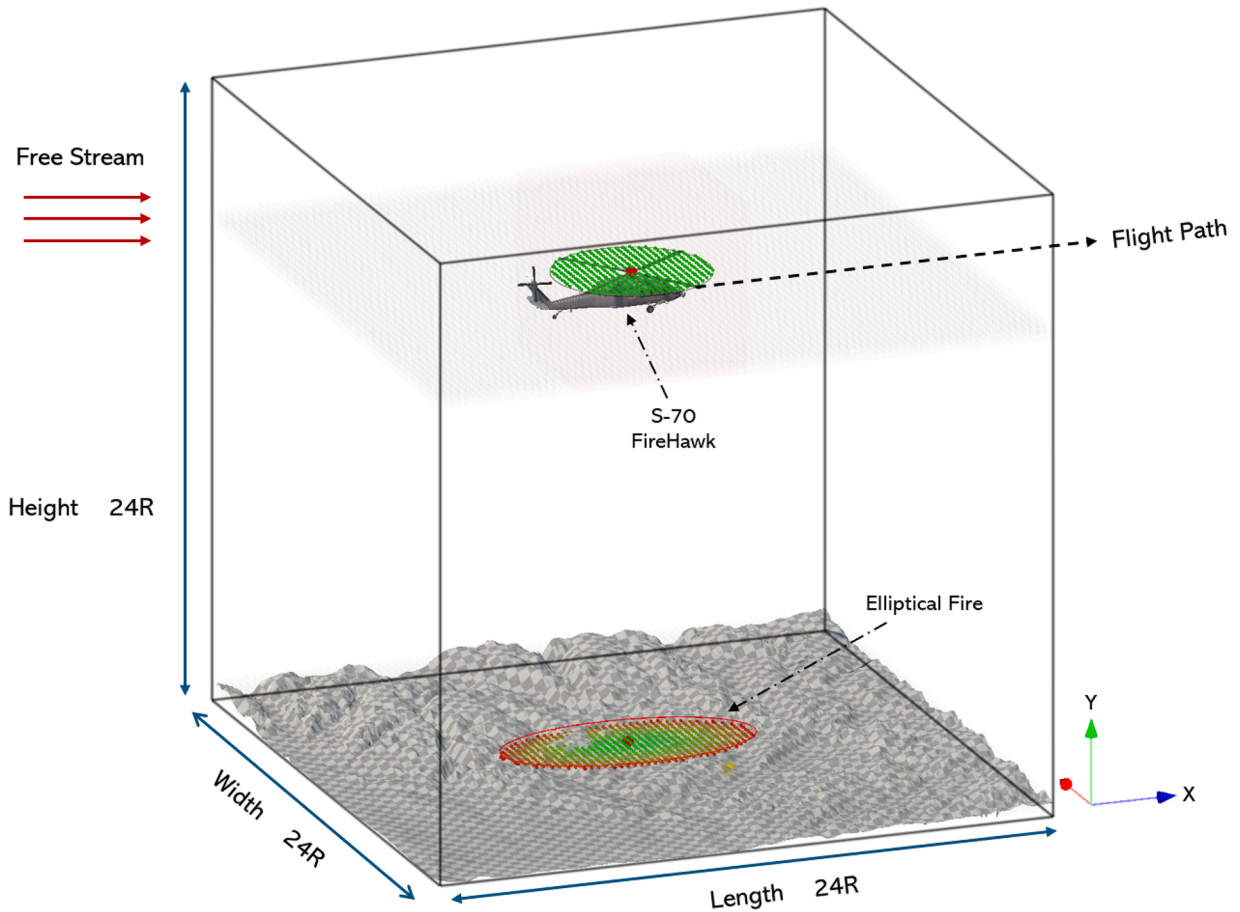


Fig. 14. Overview of the simulation domain showing the helicopter rotor, identified lattice cells within the rotor disk and the fire perimeter.

Full-coupling case where the rotor inflow was only sampled from the LBM flowfield and the computed rotor forces were fed back to the LBM solver. For the first segment of this test, a simplified fire elliptical fire model, with semi-major axis R , semi-minor axis $0.8R$, and a constant heat distribution, was used to isolate the effects of coupling, as in the full coupling mode, the helicopter downwash alters fire spread and thus heat output, introducing runtime variability between the evaluation cases.

A vertical slice through the flowfield, taken along the domain centreline with only the fire present 15, shows that the velocity induced by the buoyant plume reaches a maximum value of about 9 m/s just above the fire perimeter. However, at the altitude corresponding to the designated helicopter flight path, the upward velocity is about 6 m/s .

The recorded helicopter responses from flying over the fire are shown in Fig. 15, and two key observations are made. The first is that simply superimposing disturbances over the flight model results in an over-prediction of the effect of such disturbances. In contrast, the fully coupled system provides a more physically consistent representation of real-world expectations by capturing the mutual interactions between the aircraft and the fire. As shown in Fig. 15(d) and (g), when the fire-induced velocity is superimposed on the helicopter inflow, as done in cases 2 and 3, there is an observable reduction in the mean inflow through the helicopter rotor when flying over the plume, and an accompanying reduction in the autopilot collective command (Fig. 16).

This is because in these cases, the fire-induced velocity is simply superimposed on the local blade inflow velocity, resulting in an increase in blade angle of attack and hence thrust. As the helicopter begins to climb from this thrust increase, the autopilot issues a reduced collective command to arrest the climb and maintain the flight altitude,

lowering the computed uniform and Pitt-Peters inflow, respectively. In the fully coupled case, however, the helicopter downwash included in the loop is strong enough to reduce the effect of this upward disturbance, and thus the rotor model only sees the resultant velocity, and as can be seen from the results, the variation in inflow and control commands from the case is lower than in the one-way coupling cases.

The second observation is that the interaction between the helicopter and the fire alters the behaviours of both systems, as shown in Fig. 17. In the baseline case without the fire, the helicopter maintains a straight level path through the domain 17(a), as expected from the autopilot configuration, which maintains a constant 10 m/s forward speed, zero lateral drift, and zero climb rate. In the fully coupled case 17(b), however, the interaction between the helicopter and fire changes the behaviour of both systems. The helicopter starts a slight lateral drift to the left, while the resulting change in rotor downwash deflects the fire plume slightly to the right.

Another key result of the coupled simulation is the effect of the helicopter-induced flow on the horizontal propagation of the fire model in the fully coupled mode. By comparing the fire perimeter over the simulation runtime with and without the helicopter, shown in Fig. 18, a measurable increase in the landscape area covered by the fire can be observed as the helicopter flies over the fire region. This expansion is a result of the increased near-surface velocity around the fire perimeter due to the helicopter downwash, which increases the overall rate of spread, especially at the flank. These results demonstrate how the low-altitude helicopter operations as part of the firefighting mission can inadvertently modify the fire behaviour, and how the resulting interactions can be modelled to increase the overall fidelity of the firefighting simulation environment.

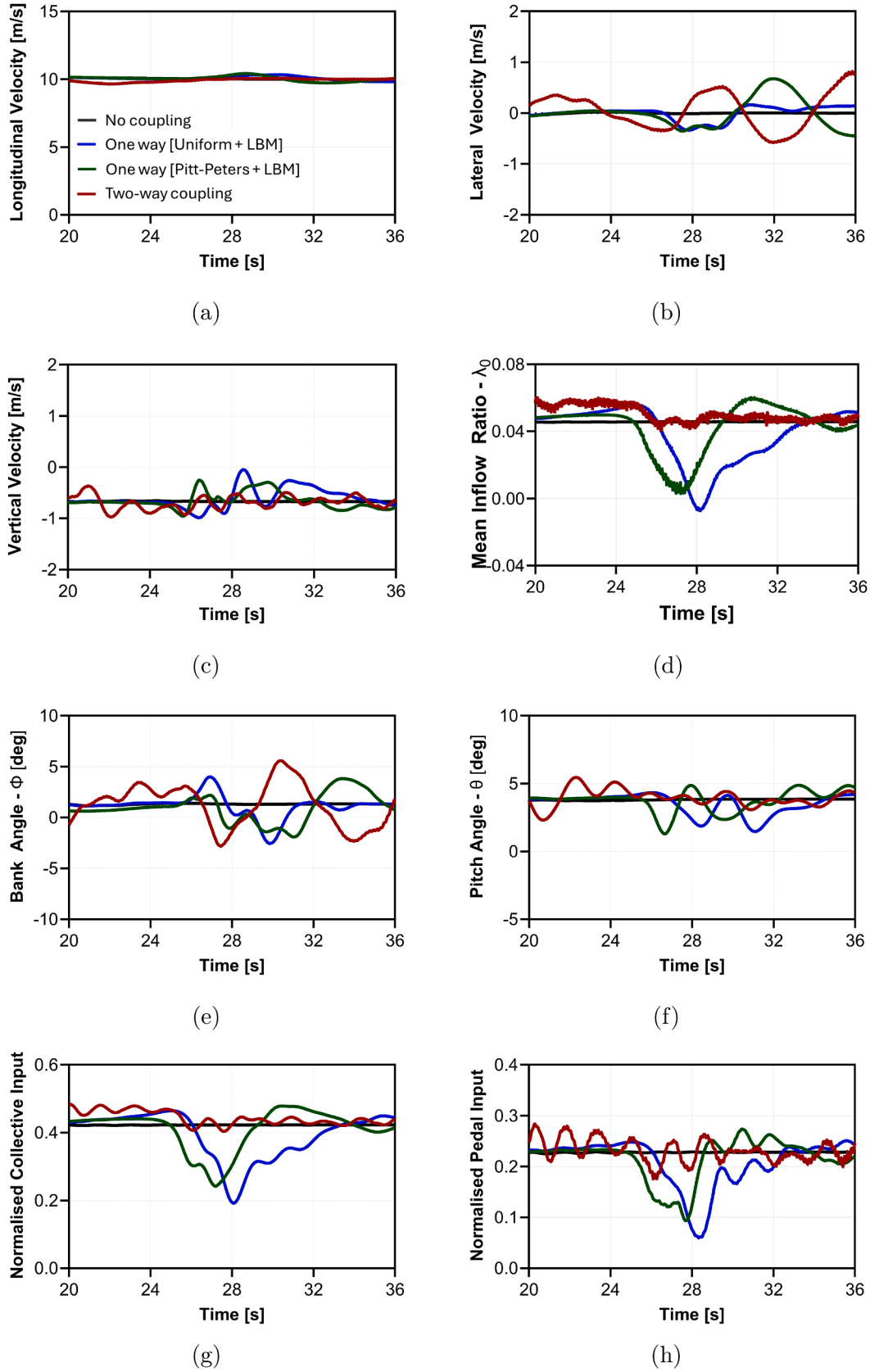


Fig. 15. Comparison between the helicopter dynamic responses across the four experimental cases. Black: No-coupling, Blue: One-way coupling A, Green: One-way coupling B and Red: Full-coupling.

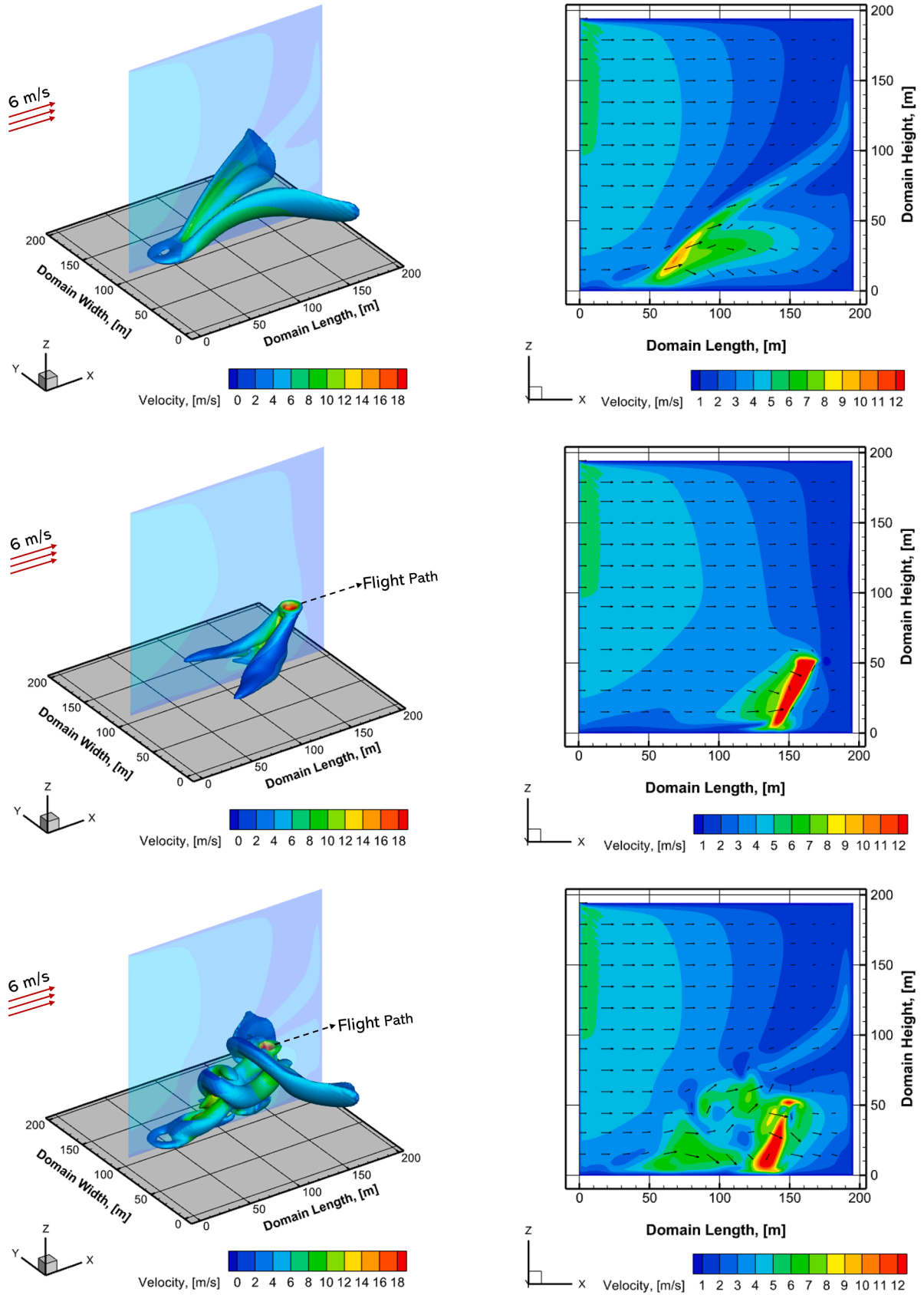


Fig. 16. Comparison of vertical slices and Q-criterion and slices of $y = 100m$, coloured with the domain velocity, between the fire only (top row), helicopter only (middle row) and the coupled simulation (bottom row) test cases.

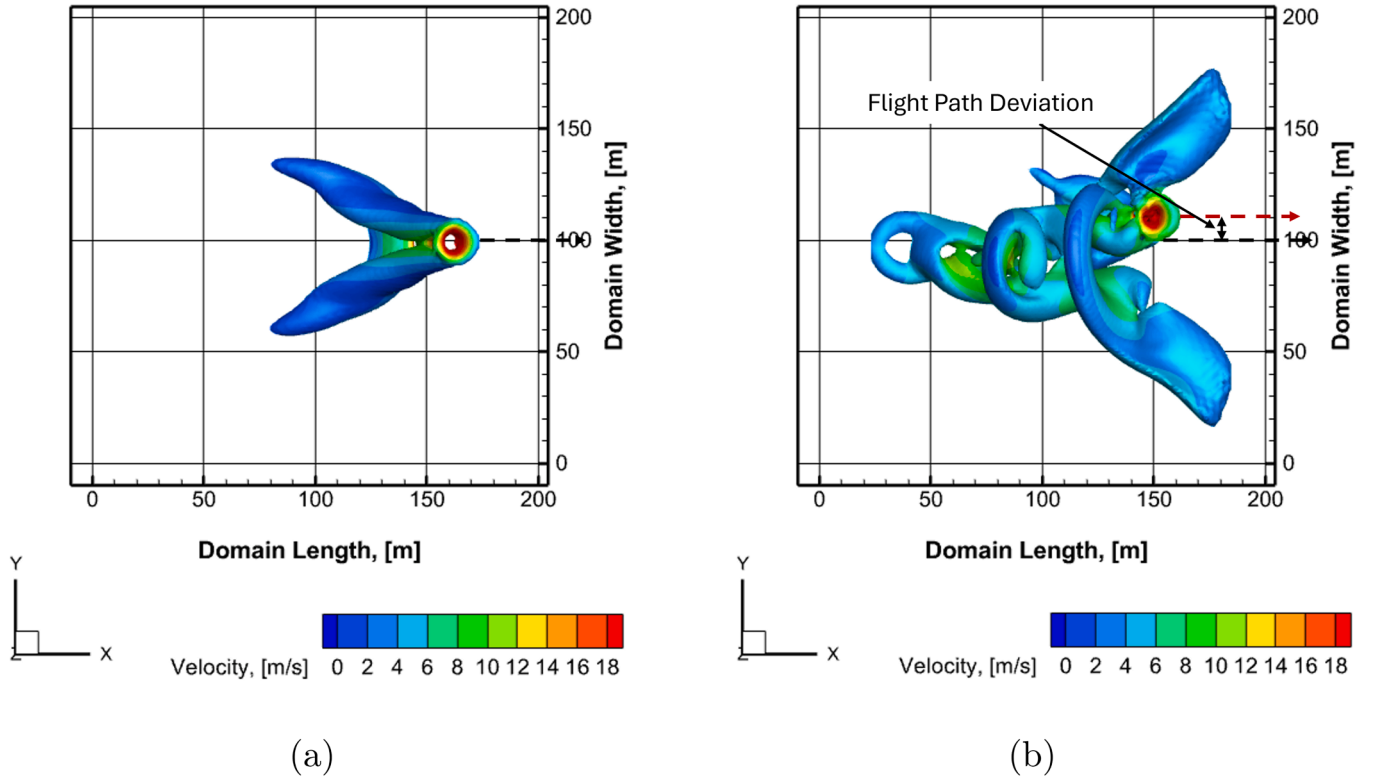


Fig. 17. An overhead view of the domain Q-criterion shaded with the measured velocity in the helicopter-only test case and the coupled fire-flight dynamics case.

3.4. Simulations of fire, terrain, water, and aircraft interactions in aerial fire fighting simulations

A second test was performed to demonstrate how the water dropped from the helicopter interacts with both the fire and the underlying terrain. This test was carried out within the same domain, but with the full fire model initialised and actively propagating across the terrain surface as shown in Fig. 18(a). In the result visualisation, the terrain properties are indicated with a gradient colour assigned to each terrain grid, with high contrast reds indicating high temperature values while high contrast blues indicate grids with high moisture content.

The SPH water model, located in a domain designed to match the helicopter position at each timestep, was initialised with 5832 particles and allowed to settle within the containing bucket bounds. The particles were then released over the fire zone, and as shown in Fig. 19(c), a noticeable drop in temperature from the terrain cells, accompanied by an increase in the ground moisture content, was observed as the first set of particles started to settle on the terrain. By the end of the 15-second simulation, about half of the landscape area that was initially burning had been extinguished, as illustrated in Fig. 19(d).

3.5. Simulation system performance evaluation

Real-time execution is crucial for the flight simulation development at hand. In this work, a target update rate of 60 frames per second (corresponding to a frame time budget of 16.67 milliseconds) was defined to ensure real-time execution and maintain proper synchronisation across the simulation subsystems, including the flight dynamics, environmental models, and especially the visual rendering plus motion cueing. This requirement is essential, as a mismatch between the cueing provided by the motion or visual system can significantly affect the flight simulation accuracy and can lead to simulator-induced motion sickness [63].

Table 2 outlines the maximum execution time allowed for each major module in milliseconds, with the indicated deviation margin subject to

Table 2

Performance budget in milliseconds for the simulation components.

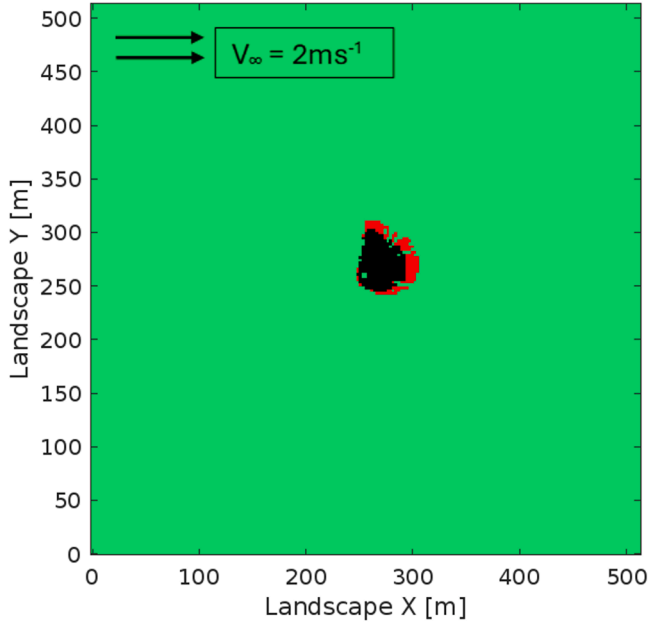
System Component	Budget (ms)
Fire Propagation	0.8 ± 0.1
Flight Model	1.0 ± 0.5
Data Transfer	3.0 ± 0.0
Atmosphere Model	5.0 ± 0.2
Water Model	2.0 ± 1.0
Rendering	4.0 ± 0.0
Total	15.3 ± 1.37

a sum constraint in that the aggregation of the extra time taken by the modules must be within the total allowed deviation.

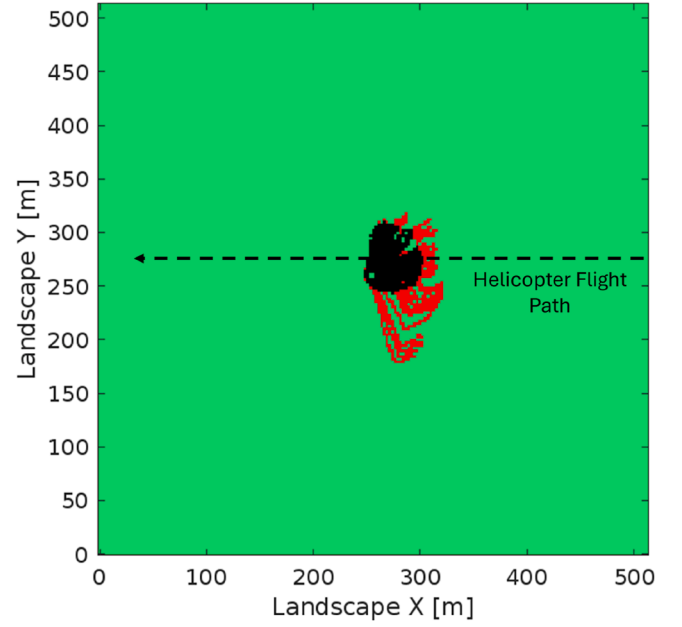
Due to the low Mach number restriction on the LBM formulation, the reference speed is low, and hence the reference time is high. This means that physical time is slower within the LBM relative to the general simulation environment. As a result, the solver must run multiple timesteps to ensure that the time is properly synchronised within the simulation. The specific number of steps needed can be calculated at each simulation step as the ratio of the physical timestep within the general simulation and the LBM, as given below:

$$t_{CR} = \frac{\Delta t_{PS_{FDM}}}{\Delta t_{PS_{LBM}}} \quad (30)$$

The performance of the developed system was evaluated on the node system with the LBM and SPH solvers running on Node 1 equipped with an Intel i9-13900K CPU (64GB RAM) and the Nvidia RTX 4090 GPU with 24 GBs of VRAM, while the other components run on Nodes 2 and 3, which are both equipped with the Intel i5-7500 CPU and Radeon RX-460 GPU. A standard metric for evaluating the computational performance of lattice Boltzmann solvers is the total number of lattice nodes, in millions, updated per second (MLUPS). This value is calculated as a

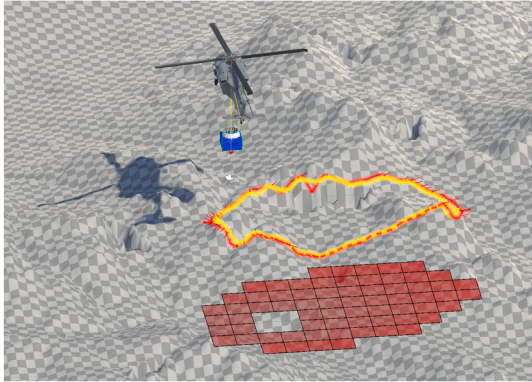


(a)

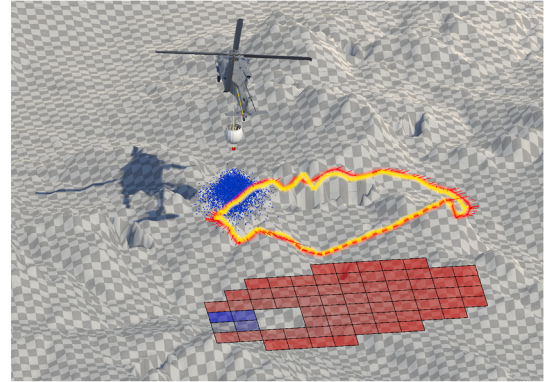


(b)

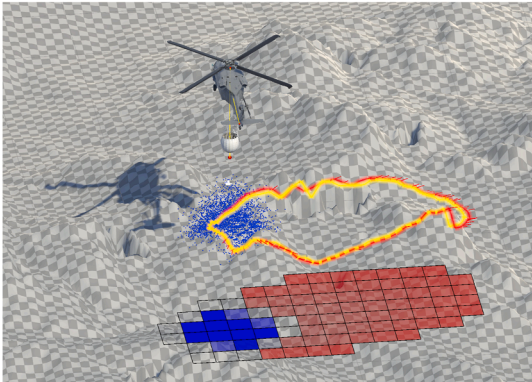
Fig. 18. Comparison between the area covered by the fire with 18(b) and without the helicopter wake interacting with the fire 18(a). Within the landscape, the grids currently on fire are indicated in red while the burnt-out regions are indicated in black.



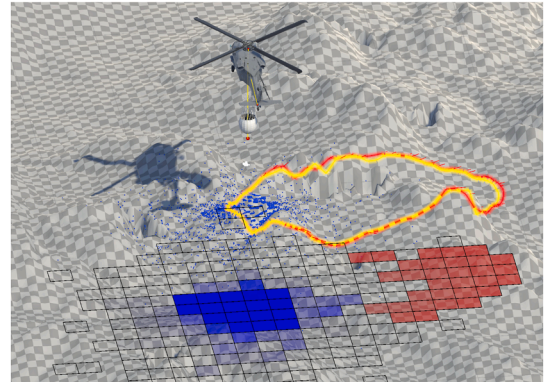
(a)



(b)



(c)



(d)

Fig. 19. Water drop demonstration showing the interaction between the fire and water particles at the drop instance 19(a), 2 secs after drop 19(b), 5 seconds after drop 19(c) and 10 seconds after drop 19(d). The moisture fraction, represented by the blue gradient, increases after the water drop, while the red gradient, representing the temperature, fades out after the water drop.

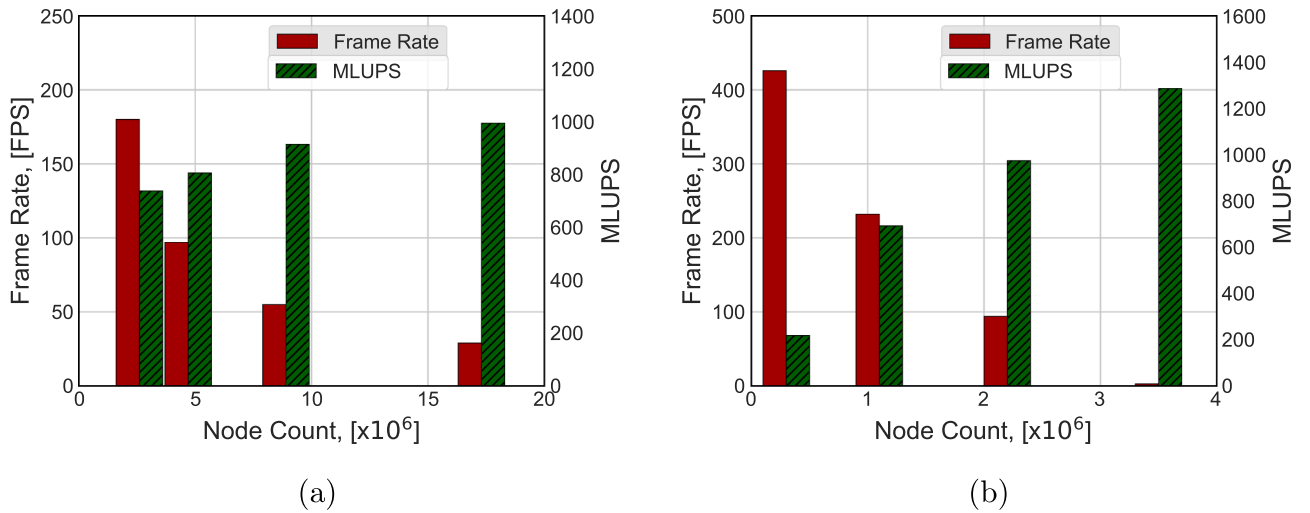


Fig. 20. Overview of the overall system performance showing the achieved frames-per-second as a function of the total node count within the LBM when the lattice timestep is free 20(a) vs when matched with the physical timestep 20(b). The total number of nodes updated per second in the solver is also indicated.

function of the lattice volume N_v given by the product of the grid sizes along the x, y, and z directions, the current number of simulation steps, C and the simulation time, T , in seconds.

$$MLUPS = \frac{N_v C}{10^6 T}. \quad (31)$$

The value achieved at various lattice nodes and the total number of simulation frames updated per second with all components running are shown in Fig. 20. The results indicate that the system meets the specified frame-per-second limit for lattice sizes up to approximately 2.1 million nodes when the lattice and physical timesteps are matched 20(a), and can handle up to 8.4 million nodes when operating in free-time mode 20(b).

4. Conclusions and future work

Previous works have explored elements of aerial firefighting simulation, but each has excluded one or more critical components. This paper presented the development of a complete, high-fidelity, fully coupled, and real-time simulation environment for aerial firefighting operations, all integrated in the DeaDalus I Flight Simulator of the University of Glasgow. The system integrates a CFD-based atmospheric model using the lattice Boltzmann method, a smoothed particle hydrodynamics-based water model, an elliptical fire propagation module, and a modular flight dynamics framework. These components were designed to run and communicate within a synchronised environment where human-in-the-loop flight simulations can be conducted. From the current results, the following conclusions can be drawn:

- Each component of the flight simulation system, i.e. the LBM-based atmospheric model, the flight model, and the fire model, was individually validated against benchmark test cases. The excellent agreements established confidence in the accuracy of each component. However, a comprehensive system-level validation is necessary, pending the availability of sufficient data from real-world firefighting operations.
- We demonstrated and evaluated the importance of system coupling for aerial firefighting simulations. Four coupling modes were tested, from a no-coupling baseline to a fully coupled configuration in which rotor forces were applied to the LBM solver and the resulting flow field directly drove the rotor inflow model. The results indicate that superimposing sampled flow disturbances onto classic inflow models (uniform or Pitt-Peters) can lead to an overprediction (for the studied cases) of induced velocity and, consequently, rotor thrust changes.

In contrast, the fully coupled mode produces rotor inflow values derived solely from the resolved LBM flow field, which inherently accounts for the feedback between the rotor wake and the surrounding environment. Secondary effects, such as crosswind-induced drift and an increase in fire spread under rotor downwash, were also naturally reproduced. These are expected in reality and were successfully captured in our fully coupled simulations.

- With a total lattice Boltzmann node count of 2.1 million running on an Nvidia 4090 GPU, the developed system was able to achieve update speeds above 90 frames a second with matched physical and lattice times, allowing for true real-time execution and coupling within the system. This was achieved via careful compute workload distribution, dynamic linking, and shared memory utilisation. This enabled real-time, high-fidelity, and fully coupled interactions between the flight model, fire and the atmosphere surrounding the simulation domain.

Future work will address current limitations in the visual representation of fire-aircraft interactions. Specifically, we will develop enhanced real-time smoke particle rendering with light scattering effects to more accurately visualise buoyant plume dynamics within the cockpit. Additionally, realistic aerial firefighting scenarios will be designed to evaluate the operational impact of water or retardant drops under various environmental and aircraft operating conditions.

CRedit authorship contribution statement

Oyedoyin Dada: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Conceptualization; **George Barakos:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Methodology, Investigation, Funding acquisition, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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